Coffee Shop Sales Analysis – 2025

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Analytical Methodology (Step-by-Step) Data Ingestion & Validation

→Load transaction dataset

• **Data Loading**

- Imported the transaction dataset into Power BI for analysis.

• **Timestamp Validation**

- Checked and converted timestamp values into the correct date and time format to ensure accurate time-based analysis.

• **Missing and Duplicate Data Check**

- Identified and removed duplicate transaction IDs.
- Checked for missing values in key fields such as Transaction ID, Date, Quantity, and Price.

• **Data Quality Verification**

- Verified that all transaction quantities and unit prices were positive values.
- Reviewed the dataset for invalid or inconsistent records and corrected or removed them where necessary.

• **Validation Confirmation**

- Confirmed that the cleaned dataset was accurate, complete, and ready for feature engineering and further analysis.

→Feature Engineering (Temporal)

• **Revenue per Transaction Calculation**

- Created a calculated column to determine the revenue generated per transaction using the formula:
  Revenue per Transaction = Transaction Quantity × Unit Price

- **Hour Extraction**
    - Extracted the hour (0–23) from the transaction timestamp to analyze hourly sales patterns.
- **Day of Week Extraction**
    - Derived the day of the week (Monday–Sunday) from the transaction date to compare sales performance across different weekdays.
- **Time Bucket Creation**
    - Classified each transaction into predefined time buckets based on the hour of purchase:
        - **Morning:** 6:00 AM – 11:59 AM
        - **Afternoon:** 12:00 PM – 4:59 PM
        - **Evening:** 5:00 PM – 9:59 PM
        - **Late Hours:** 10:00 PM – 5:59 AM
- **Validation**
    - Verified that all engineered features were correctly categorized and ready for use in time-based analysis and visualizations.

→Sales Trend Analysis

- **Data Collection**
    - Imported the Afficionado Coffee Roasters sales dataset into Power BI.
- **Data Cleaning and Preparation**
    - Checked for missing values and duplicates.
    - Converted the date column into the correct date format.
    - Verified sales and transaction data for accuracy.
- **Data Modeling**
    - Created relationships between tables (if required).
    - Developed DAX measures such as **Total Revenue** and **Total Transactions**.
- **Daily Revenue Trend Analysis**
    - Used a **Line Chart** to analyze daily revenue trends.
    - Identified days with high and low sales performance.

- **Weekly Revenue and Transaction Analysis**
  - Aggregated revenue and transaction data on a weekly basis.
  - Compared weekly performance using column or line charts.
- **Trend Pattern Identification**
  - Examined sales data to identify upward and downward trends over time.
  - Observed seasonal or recurring sales patterns.
- **Store-Level Trend Comparison**
  - Compared revenue trends across different store locations.
  - Identified the best-performing and lowest-performing stores.
- **Insights and Recommendations**
  - Summarized key findings from the analysis.
  - Suggested actions to improve sales performance and operational planning.

→Day-of-Week Performance Analysis

- **Data Loading**
  - Imported the transaction dataset into Power BI for analysis.
- **Timestamp Validation**
  - Checked and converted timestamp values into the correct date and time format to ensure accurate time-based analysis.
- **Missing and Duplicate Data Check**
  - Identified and removed duplicate transaction IDs.
  - Checked for missing values in key fields such as Transaction ID, Date, Quantity, and Price.
- **Data Quality Verification**
  - Verified that all transaction quantities and unit prices were positive values.
  - Reviewed the dataset for invalid or inconsistent records and corrected or removed them where necessary.
- **Validation Confirmation**
  - Confirmed that the cleaned dataset was accurate, complete, and ready for feature engineering and further analysis.

→Time-of-Day Demand Analysis

• **Hourly Transaction Analysis**

- Grouped transactions by hour of the day (0–23) to analyze hourly transaction volume.

• **Hourly Revenue Analysis**

- Calculated total revenue for each hour to understand the distribution of sales throughout the day.

• **Visualization**

- Created line and column charts in Power BI to display hourly transaction volume and hourly revenue trends.

• **Peak and Slow Period Identification**

- Identified **morning rush hours** by analyzing periods with the highest transaction volume.
- Detected **midday slow periods** where both revenue and transaction counts were relatively low.
- Identified **evening peak hours** by observing the highest revenue and customer activity during evening time.

• **Insights and Recommendations**

- Summarized key hourly demand patterns.
- Provided recommendations for staff scheduling and operational planning based on peak and slow business hours.

→Cross-Location Temporal Comparison

• **Store-wise Data Segmentation**

- Segmented the dataset by store location to analyze sales performance for each store independently.

• **Hourly Heatmap Creation**

- Created hourly heatmaps in Power BI to visualize transaction volume and revenue across different hours for each store.

• **Peak-Hour Comparison**

- Compared peak business hours across all store locations.
- Identified whether stores showed similar peak-hour patterns (alignment) or different demand patterns (divergence).

- **Location-Specific Customer Behaviour Analysis**

    - Analyzed customer purchasing behaviour at each location based on hourly transaction activity and revenue.
    - Identified unique demand patterns and customer preferences for individual stores.

- **Insights and Recommendations**

    - Summarized key differences and similarities in temporal sales trends across locations.
    - Recommended location-specific staffing, inventory planning, and operational strategies based on customer behaviour and peak and demand periods.